

Testing the equality of two high-dimensional spatial sign covariance matrices

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Abstract

This paper is concerned with testing the equality of two high-dimensional spatial sign covariance matrices with applications to testing the proportionality of two high-dimensional covariance matrices. It is interesting that these two testing problems are completely equivalent for the class of elliptically symmetric distributions. This paper develops a new test for testing the equality of two high-dimensional spatial sign covariance matrices based on the Frobenius norm of the difference between two spatial sign covariance matrices. The asymptotic normality of the proposed testing statistic is derived under the null and alternative hypotheses when the dimension and sample sizes both tend to infinity. Moreover, the asymptotic power function is also presented. Simulation studies show that the proposed test performs very well in a wide range of settings and can be allowed for the case of large dimensions and small sample sizes.

KEYWORDS

elliptically symmetric distribution, high dimension, spatial sign covariance matrix, U-statistic

1 | INTRODUCTION

Consider two populations $\mathbf{X}$ and $\mathbf{Y}$ with p -dimensional mean vectors μ_1 and μ_2 and $p \times p$ -dimensional covariance matrices Σ_1 and Σ_2 , respectively. Much interest has been focused on learning the relationship between Σ_1 and Σ_2 . One of them is to test the following hypothesis:

$$H_0 : \Sigma_1 = c\Sigma_2 \quad \text{versus} \quad H_1 : \Sigma_1 \neq c\Sigma_2, \quad (1)$$

for some unknown scalar $c > 0$. It is interesting that for elliptically symmetric distributions, testing the proportionality $\Sigma_1 = c\Sigma_2$ is completely equivalent to testing $\mathbf{S}_1 = \mathbf{S}_2$, where $\mathbf{S}_1 = E[\|\mathbf{X} - \mu_1\|^{-2}(\mathbf{X} - \mu_1)(\mathbf{X} - \mu_1)^T]$ and $\mathbf{S}_2 = E[\|\mathbf{Y} - \mu_2\|^{-2}(\mathbf{Y} - \mu_2)(\mathbf{Y} - \mu_2)^T]$ are two spatial sign covariance matrices with $\|\cdot\|$ denoting the L_2 -norm. The equivalence will be proved in Section 2.2. The target of this paper is to develop a novel nonparametric procedure to test

$$H'_0 : \mathbf{S}_1 = \mathbf{S}_2 \quad \text{versus} \quad H'_1 : \mathbf{S}_1 \neq \mathbf{S}_2, \quad (2)$$

with applications to test the proportionality $H_0 : \Sigma_1 = c\Sigma_2$ for elliptically symmetric distributions.

The literature about testing the equality of high-dimensional spatial sign covariance matrices is sparse, but there is a rich literature on testing the proportionality of two high-dimensional covariance matrices. The proportionality test of (1) is widely used in various areas, for example, discriminant analysis and principal component analysis (Flury & Riedwyl, 1988; Schott, 1991), and it is also very attractive in the scientific fields of economics and genetics (Flury, 1986; Jensen & Madsen, 2004). Since Federer (1951) first studied the proportional relationship of two covariance matrices with dimension $p \leq 3$, a number of methods have been developed on the proportionality test of covariance matrices (for example, Kim, 1971; Rao, 1983; Flury, 1986; Eriksen, 1987; Schott, 1999). However, all of these tests are based on the classical limit theorem, which assumes that the sample size tends to infinity but the dimension p is fixed. With the rapid development and wide applications of computer techniques, dimension p is no longer very small with respect to the sample size and even diverges to infinity as the sample size goes to infinity. This often leads to the classical tests performing poorly, being unstable for large p values and even being unavailable when p is larger than $n_1 + n_2$ (Bai & Saranadasa, 1996). Thus, it is very urgent to develop some new methods to deal with the proportionality test of (1) for high-dimensional data. Assuming that dimension p increases proportionally with the sample size n_l , that is, $p/n_l \rightarrow q_l \in (0, 1)$, $l = 1, 2$, Xu, Liu, Zheng, and Bao (2014) developed a pseudo-likelihood ratio test (PLRT), which extended the traditional likelihood ratio test, and established its asymptotic normality under the existence of finite fourth population moments. Moreover, Liu, Xu, Zheng, and Tian (2014) proposed a method that allowed $p/n_1 \rightarrow q_1 \in (0, \infty)$ and showed that the test statistic satisfied asymptotic normality. These methods are summarized in Table 1.

TABLE 1 Reviews of the developed tests for the proportionality of covariance matrices on different population distributions

Method	p	Distributions
Flury (1986)	fixed	two multivariate normal populations
Eriksen (1987)	fixed	$k \geq 2$ multivariate normal populations
Schott (1999)	fixed	$k \geq 2$ populations with finite fourth moments
Xu et al. (2014)	$p/n_l \in (0, 1)$, $l = 1, 2$	two populations with finite fourth moments
Liu et al. (2014)	$p/n_1 \in (0, \infty)$, $p/n_2 \in (0, 1)$	two populations with finite fourth moments

Note. n_1 and n_2 are sample sizes of two populations.

A shared drawback of the methods by Xu et al. (2014) and Liu et al. (2014) is that both have unsatisfactory size and power performance when the population distribution is heavy tailed (see Table 3 in Section 4). Moreover, when $p > \max(n_1, n_2)$, they cannot be defined due to the nonexistence of the inverse of the sample covariance matrix.

In a multivariate data analysis, it is often assumed that the observations are from a Gaussian population. However, it is well known that, in real data analysis, the observed data seldom follow a Gaussian distribution but often display some non-normal features such as heavy tails, skewness, etc. For example, it is observed that the marginal distributions of the microarray expressions are non-normal even after log transformation and have heavy tails based on values of their marginal kurtosis in the gene expression data (Purdom & Holmes, 2005; Wang, Peng, & Li, 2015). Furthermore, “outliers in microarray expression data frequently arise due to the array chip artifacts such as uneven spray of reagents within arrays” (Wang et al., 2015). A good alternative approach is to consider a more flexible class of distributions such as a multivariate t -distribution, which has been shown to be more robust than a multivariate normal distribution. This motivates us to consider a class of elliptically symmetric distributions, which includes a multivariate Gaussian distribution, a multivariate t -distribution, a multivariate logistic distribution, and a Pearson Type II multivariate distribution as special ones.

In this paper, we propose a new high-dimensional test for hypothesis (2) based on sample spatial sign covariance matrices. On the one hand, the proposed test can also be used to test hypothesis (1) for elliptically symmetric distributions. On the other hand, the proposed test is robust against heavy-tailed data and can be allowed to the case of $p > \max(n_1, n_2)$. The proposed testing statistic is asymptotically normal under the null and alternative hypotheses. Simulation studies show that our procedure can keep the empirical sizes well at the nominal size and achieve great empirical power for a wide range of dimensions, sample sizes, and population distributions.

The rest of this paper is organized as follows. We first prove the equivalence between testing the equality of two spatial sign covariance matrices and testing the proportionality of two covariance matrices for elliptically symmetric distributions in Section 2. Section 3 introduces our high-dimensional nonparametric test statistic and its limiting distributions under the null and alternative hypotheses. Monte Carlo simulation studies are conducted in Section 4. Section 5 includes conclusions and discussions. Technical proofs are given in the Appendix.

2 | EQUIVALENCE OF EQUALITY OF TWO SPATIAL SIGN COVARIANCE MATRICES AND PROPORTIONALITY OF TWO COVARIANCE MATRICES FOR ELLIPTICALLY SYMMETRIC DISTRIBUTIONS

2.1 | Review of elliptically symmetric distributions

In this section, we briefly review some theories of elliptically symmetric distributions. A p -dimensional random vector $\mathbf{X}$ is said to follow an elliptically symmetric distribution if it has the following stochastic representation (Fang, Kotz, & Ng, 1990):

$$\mathbf{X} = \boldsymbol{\mu} + \xi \boldsymbol{\Gamma} \mathbf{W},$$

where $\boldsymbol{\mu}$ is the mean vector, $\boldsymbol{\Gamma}$ is a nonrandom and invertible $p \times p$ -dimensional matrix, ξ is a univariate nonnegative random variable, and $\mathbf{W}$ is a p -dimensional random vector from a uniform distribution on the unit sphere in R^p , with ξ and $\mathbf{W}$ being independent, $E(\mathbf{W}\mathbf{W}^T) = p^{-1}\mathbf{I}_p$, and

$\mathbf{I}_p$ is the $p \times p$ -dimensional identity matrix. The covariance matrix Σ and shape matrix Λ of the elliptical symmetric population $\mathbf{X}$ satisfy

$$\Sigma = p^{-1}E(\xi^2) \cdot \Lambda, \quad \Lambda = \Gamma\Gamma^T, \quad \Sigma \propto \Lambda,$$

where A^T denotes the transpose of a vector or a matrix A . The class of elliptical distributions includes many useful multivariate distributions, such as the multivariate Gaussian distribution, the multivariate t -distribution, the multivariate logistic distribution, the Pearson Type II multivariate distribution, and so on.

2.2 | Equivalence of two testing hypotheses (1) and (2) for elliptically symmetric distributions

Suppose that $\mathbf{X}$ and $\mathbf{Y}$ are from two p -dimensional elliptically symmetric populations with mean vectors μ_1 and μ_2 and covariance matrices Σ_1 and Σ_2 , respectively. The samples $\mathbf{x}_1, \dots, \mathbf{x}_{n_1}$ are independent and identically distributed from $\mathbf{X}$, and the samples $\mathbf{y}_1, \dots, \mathbf{y}_{n_2}$ are independent and identically distributed from $\mathbf{Y}$. The shape matrices are $\Lambda_l = c_{lp}^{-1}\Sigma_l$ for some scalar $c_{lp} > 0, l = 1, 2$. Without loss of generality, assume that $\text{tr}(\Lambda_1) = \text{tr}(\Lambda_2) = p$, where $\text{tr}(\cdot)$ denotes the trace of a square matrix. For a given vector $\mathbf{x}$, the corresponding multivariate spatial sign function is defined by $U(\mathbf{x}) = \|\mathbf{x}\|^{-1}\mathbf{x}$, where $\|\mathbf{x}\|$ is the L_2 -norm of $\mathbf{x}$. Let

$$\mathbf{u}_i = U(\mathbf{x}_i - \mu_1) \text{ and } \mathbf{v}_j = U(\mathbf{y}_j - \mu_2), i = 1, \dots, n_1, j = 1, \dots, n_2,$$

be the spatial sign functions of the centralized $\mathbf{x}_i$ and $\mathbf{y}_j$, respectively. When the center parameters μ_1 and μ_2 are known, the corresponding sample spatial sign covariance matrices are defined as

$$\mathbf{S}_{1n_1} = n_1^{-1} \sum_{i=1}^{n_1} \mathbf{u}_i \mathbf{u}_i^T \text{ and } \mathbf{S}_{2n_2} = n_2^{-1} \sum_{j=1}^{n_2} \mathbf{v}_j \mathbf{v}_j^T,$$

respectively, with $\mathbf{S}_l = E(\mathbf{S}_{ln_l}), l = 1, 2$. About the relationship of Σ_l and $\mathbf{S}_l$, there is a very useful lemma from Magyar and Tyler (2014).

Lemma 1. (See equation 3.9 of Magyar & Tyler, 2014) For the p -dimensional elliptically symmetric population $\mathbf{X}$, let $\lambda_1 \geq \dots \geq \lambda_p$ be the eigenvalues of Σ_1 and $\psi_1 \geq \dots \geq \psi_p$ be the eigenvalues of $\mathbf{S}_1$. Then, we have

$$\psi_h = E \left(\lambda_h \chi_{1,h}^2 / \sum_{r=1}^p \lambda_r \chi_{1,r}^2 \right), \quad h = 1, \dots, p,$$

where $\chi_{1,1}^2, \dots, \chi_{1,p}^2$ are mutually independent χ^2 -variables with one degree of freedom 1.

Lemma 1 shows that when the eigenvalues of the covariance matrices Σ_1 and Σ_2 are proportional, the spatial sign covariance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ have the same eigenvalues. Similarly, we are fortunate to obtain that when the spatial sign covariance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ have the same eigenvalues, the eigenvalues of the covariance matrices Σ_1 and Σ_2 are proportional.

Theorem 1. For two p -dimensional elliptically symmetric populations $\mathbf{X}$ and $\mathbf{Y}$, if $\psi_h = \psi'_h, h = 1, \dots, p$, then we have

$$\lambda_1 / \lambda'_1 = \dots = \lambda_p / \lambda'_p,$$

where $\psi_1 \geq \dots \geq \psi_p$ and $\lambda_1 \geq \dots \geq \lambda_p$ are the eigenvalues of $\mathbf{S}_1$ and Σ_1 , and $\psi'_1 \geq \dots \geq \psi'_p$ and $\lambda'_1 \geq \dots \geq \lambda'_p$ are the eigenvalues of $\mathbf{S}_2$ and Σ_2 , respectively.

The proof of Theorem 1 is placed in the Appendix.

From section 3.1 of Magyar and Tyler (2014), we know that Σ_l and S_l have the same eigenvectors, $l = 1, 2$, for elliptically symmetric populations. Then, by Lemma 1 and Theorem 1, we know that the null hypothesis (1) is completely equivalent to the null hypothesis $H'_0 : S_1 = S_2$, that is,

$$H_0 : \Sigma_1 = c\Sigma_2 \iff H'_0 : S_1 = S_2.$$

Thus, testing the proportionality of two covariance matrices Σ_1 and Σ_2 is equivalent to testing the equality of two spatial sign covariance matrices S_1 and S_2 for elliptically symmetric distributions.

3 | TESTING THE EQUALITY OF TWO HIGH-DIMENSIONAL SPATIAL SIGN COVARIANCE MATRICES FOR ELLIPTICALLY SYMMETRIC DISTRIBUTIONS

To test the equality of two high-dimensional spatial sign covariance matrices, we require two assumptions.

Assumption 1. Assume that the moments $E(R_{1i}^{-4})$ and $E(R_{2j}^{-4})$ exist for $1 \leq i \leq n_1, 1 \leq j \leq n_2$ with $R_{1i} = \|\mathbf{x}_i - \mu_1\|$ and $R_{2j} = \|\mathbf{y}_j - \mu_2\|$. Moreover, for $2 \leq k \leq 4$, we have $E(R_{1i}^{-k})/E^k(R_{1i}^{-1}) \rightarrow d_{1k} \in [1, +\infty)$ and $E(R_{2j}^{-k})/E^k(R_{2j}^{-1}) \rightarrow d_{2k} \in [1, +\infty)$ as $p \rightarrow \infty$, where d_{1k} and d_{2k} are constants.

Assumption 2. Assume $n_l/(n_1 + n_2) \rightarrow \eta_l \in (0, 1)$ as $\min\{n_1, n_2\} \rightarrow \infty$ and $p = O(n_l^2)$, for $l = 1, 2$. Moreover, as $p \rightarrow \infty$, assume $\text{tr}(\Sigma_{l_1} \Sigma_{l_2} \Sigma_{l_3} \Sigma_{l_4}) = o(\text{tr}(\Sigma_{l_1} \Sigma_{l_2}) \text{tr}(\Sigma_{l_3} \Sigma_{l_4}))$ for $l_1, l_2, l_3, l_4 \in \{1, 2\}$.

For simplicity, we let $n = n_1 + n_2$. Similar to Zou, Peng, Feng, and Wang (2014), Assumption 1 is a necessary condition to ensure the validity of the second-order expansions. Assumption 2 is similar to Li and Chen (2012). When all the eigenvalues of both Σ_1 and Σ_2 are bounded, Assumption 2 holds. However, for the matrix having the maximum eigenvalue being $O(p)$, for instance, the compound symmetric matrix, Assumption 2 does not hold. Moreover, Assumption 2 requires that the convergence order between dimension and sample sizes, that is, $p = O(n_l^2)$ for $l = 1, 2$.

3.1 | Testing statistic

The Frobenius norm for any $m \times n$ matrix D is defined as $\|D\|_F^2 = \text{tr}(D^T D)$. It is known that S_{ln_l} is a sample version of S_l for $l = 1, 2$. This motivates us to construct a random variable to measure the difference between S_{1n_1} and S_{2n_2} under the Frobenius norm as follows:

$$\begin{aligned} T_{n_1, n_2}^* &= p \text{tr}(S_{1n_1} - S_{2n_2})^2 \\ &= p \left\{ n_1^{-1} + n_1^{-2} \sum_{i \neq i'}^{n_1} (\mathbf{u}_i^T \mathbf{u}_{i'})^2 + n_2^{-1} + n_2^{-2} \sum_{j \neq j'}^{n_2} (\mathbf{v}_j^T \mathbf{v}_{j'})^2 - 2n_1^{-1} n_2^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\mathbf{u}_i^T \mathbf{v}_j)^2 \right\}. \end{aligned}$$

Ignoring the nonrandom terms n_1^{-1}, n_2^{-1} , using $\{n_1(n_1 - 1)\}^{-1}, \{n_2(n_2 - 1)\}^{-1}$ as a multiplier instead of n_1^{-2} and n_2^{-2} in the first two random terms, we arrive at a modified version of T_{n_1, n_2}^* as

$T_{n_1, n_2} = p(A_{n_1} + B_{n_2} - 2C_{n_1, n_2})$, where

$$A_{n_1} = \frac{1}{n_1(n_1 - 1)} \sum_{i \neq i'}^{n_1} (\mathbf{u}_i^T \mathbf{u}_{i'})^2, \quad B_{n_2} = \frac{1}{n_2(n_2 - 1)} \sum_{j \neq j'}^{n_2} (\mathbf{v}_j^T \mathbf{v}_{j'})^2, \quad C_{n_1, n_2} = \frac{1}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\mathbf{u}_i^T \mathbf{v}_j)^2. \quad (3)$$

It is clear that $E(T_{n_1, n_2}) = p \operatorname{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$. In practice, μ_1 and μ_2 are unknown. By replacing μ_1 and μ_2 with the spatial median estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ in T_{n_1, n_2} (Mottonen & Oja, 1995), we propose the statistic for testing the equality of two spatial sign covariance matrices $\mathbf{S}_1$ and $\mathbf{S}_2$ as

$$T'_{n_1, n_2} = \frac{p}{n_1(n_1 - 1)} \sum_{i \neq i'}^{n_1} (\hat{\mathbf{u}}_i^T \hat{\mathbf{u}}_{i'})^2 + \frac{p}{n_2(n_2 - 1)} \sum_{j \neq j'}^{n_2} (\hat{\mathbf{v}}_j^T \hat{\mathbf{v}}_{j'})^2 - \frac{2p}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} (\hat{\mathbf{u}}_i^T \hat{\mathbf{v}}_j)^2,$$

where $\hat{\mathbf{u}}_i = U(\mathbf{x}_i - \hat{\mu}_1)$, $i = 1, \dots, n_1$ and $\hat{\mathbf{v}}_j = U(\mathbf{y}_j - \hat{\mu}_2)$, $j = 1, \dots, n_2$. Consider $\mathbf{r}_i = B\mathbf{x}_i$, $\mathbf{w}_j = B\mathbf{y}_j$, where B is an orthogonal matrix. Note that the spatial median estimators $\hat{\mu}_1$ and $\hat{\mu}_2$ are orthogonal invariant, $\hat{\mu}_r = B\hat{\mu}_1$, $\hat{\mu}_w = B\hat{\mu}_2$. Then, we have

$$\hat{\mathbf{u}}_{\mathbf{r}_i} = U(\mathbf{r}_i - \hat{\mu}_r) = BU(\mathbf{x}_i - \hat{\mu}_1) = B\hat{\mathbf{u}}_i,$$

and $(\hat{\mathbf{u}}_{\mathbf{r}_i}^T \hat{\mathbf{u}}_{\mathbf{r}_{j'}})^2 = (\hat{\mathbf{u}}_i^T B^T B \hat{\mathbf{u}}_{j'})^2 = (\hat{\mathbf{u}}_i^T \hat{\mathbf{u}}_{j'})^2$, $(\hat{\mathbf{v}}_{\mathbf{w}_j}^T \hat{\mathbf{v}}_{\mathbf{w}_{j'}})^2 = (\hat{\mathbf{v}}_j^T \hat{\mathbf{v}}_{j'})^2$, $(\hat{\mathbf{u}}_{\mathbf{r}_i}^T \hat{\mathbf{v}}_{\mathbf{w}_j})^2 = (\hat{\mathbf{u}}_i^T \hat{\mathbf{v}}_j)^2$. Hence, T'_{n_1, n_2} is also orthogonal invariant when $\mathbf{x}_i$ and $\mathbf{y}_j$ take the same orthogonal transformations.

3.2 | Limiting distributions of the testing statistic

To consider the bias of the testing statistic T'_{n_1, n_2} , the following lemma will first give the expectation of T'_{n_1, n_2} .

Lemma 2. Under Assumptions 1 and 2 and elliptically symmetric distributions, as $\min\{n_1, n_2\} \rightarrow \infty$ and $p \rightarrow \infty$, we have

$$E(T'_{n_1, n_2}) = p \operatorname{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2 + p\delta_{n_1, n_2}, \quad (4)$$

where

$$\begin{aligned} \delta_{n_1, n_2} = & n_1^{-2} \left\{ 2 - \frac{2E(R_{11}^{-2})}{E^2(R_{11}^{-1})} + \frac{E^2(R_{11}^{-2})}{E^4(R_{11}^{-1})} \right\} + n_2^{-2} \left\{ 2 - \frac{2E(R_{22}^{-2})}{E^2(R_{21}^{-1})} + \frac{E^2(R_{21}^{-2})}{E^4(R_{21}^{-1})} \right\} \\ & + n_1^{-3} \left\{ -\frac{6E^2(R_{11}^{-2})}{E^4(R_{11}^{-1})} + \frac{2E(R_{11}^{-2})E(R_{11}^{-3})}{E^5(R_{11}^{-1})} + \frac{8E(R_{11}^{-2})}{E^2(R_{11}^{-1})} - \frac{2E(R_{11}^{-3})}{E^3(R_{11}^{-1})} \right\} \\ & + n_2^{-3} \left\{ -\frac{6E^2(R_{21}^{-2})}{E^4(R_{21}^{-1})} + \frac{2E(R_{21}^{-2})E(R_{21}^{-3})}{E^5(R_{21}^{-1})} + \frac{8E(R_{21}^{-2})}{E^2(R_{21}^{-1})} - \frac{2E(R_{21}^{-3})}{E^3(R_{21}^{-1})} \right\}. \end{aligned} \quad (5)$$

The proof of Lemma 2 is provided in the Supplementary Material.

Then, we obtain the central limit theorem of the proposed testing statistic under both null and alternative hypotheses in the following theorem.

Theorem 2. Under Assumptions 1 and 2 and elliptically symmetric distributions, as $\min\{n_1, n_2\} \rightarrow \infty$ and $p \rightarrow \infty$, we have the following.

1. $v_{n_1, n_2}^{-1} \{T'_{n_1, n_2} - p\delta_{n_1, n_2} - p \operatorname{tr}[(\mathbf{S}_1 - \mathbf{S}_2)^2]\} \xrightarrow{D} N(0, 1)$, where $\xrightarrow{D}$ denotes convergence in distribution and

$$\begin{aligned} v_{n_1, n_2}^2 = & \frac{4}{n_1(n_1 - 1)} \frac{\operatorname{tr}^2(\Lambda_1^2)}{(p + 2)^2} + \frac{8}{n_1} \frac{p \operatorname{tr}(\Lambda_1^4) - \operatorname{tr}^2(\Lambda_1^2)}{p^2(p + 2)} \\ & + \frac{4}{n_2(n_2 - 1)} \frac{\operatorname{tr}^2(\Lambda_2^2)}{(p + 2)^2} + \frac{8}{n_2} \frac{p \operatorname{tr}(\Lambda_2^4) - \operatorname{tr}^2(\Lambda_2^2)}{p^2(p + 2)} \\ & + \frac{8}{n_1 n_2} \frac{\operatorname{tr}^2(\Lambda_1 \Lambda_2)}{(p + 2)^2} + \left(\frac{8}{n_1} + \frac{8}{n_2} \right) \frac{p \operatorname{tr}(\Lambda_1 \Lambda_2)^2 - \operatorname{tr}^2(\Lambda_1 \Lambda_2)}{p^2(p + 2)} \\ & - \frac{16}{n_1} \frac{p \operatorname{tr}(\Lambda_1^3 \Lambda_2) - \operatorname{tr}(\Lambda_1 \Lambda_2) \operatorname{tr}(\Lambda_1^2)}{p^2(p + 2)} - \frac{16}{n_2} \frac{p \operatorname{tr}(\Lambda_2^3 \Lambda_1) - \operatorname{tr}(\Lambda_1 \Lambda_2) \operatorname{tr}(\Lambda_2^2)}{p^2(p + 2)}. \end{aligned}$$

2. Under $H_0 : \mathbf{S}_1 = \mathbf{S}_2$, we have $v_{0, n_1, n_2}^{-1} \{T'_{n_1, n_2} - p\delta_{n_1, n_2}\} \xrightarrow{D} N(0, 1)$, where δ_{n_1, n_2} is in (5), and $v_{0, n_1, n_2}^2 = 4(n_1^{-1} + n_2^{-1})^2(p + 2)^{-2} \operatorname{tr}^2(\Lambda^2)$ with $\Lambda = \Lambda_1 = \Lambda_2$ under H_0 .

Remark 1. To formulate our testing procedure, we need to estimate the unknown quantities in δ_{n_1, n_2} and v_{0, n_1, n_2} under $H_0 : \mathbf{S}_1 = \mathbf{S}_2$. The moment estimate of $E(R_{li}^{-k})$ is $\hat{E}(R_{li}^{-k}) = n_l^{-1} \sum_{i=1}^{n_l} \hat{R}_{li}^{-k}$ for $l = 1, 2$ with $\hat{R}_{li} = \|\mathbf{x}_i - \hat{\boldsymbol{\mu}}_1\|$ and $\hat{R}_{2i} = \|\mathbf{y}_i - \hat{\boldsymbol{\mu}}_2\|$. In the Supplementary Material (p. 6), we proved that

$$p \operatorname{tr}(\mathbf{S}_l^2) = p^{-1} \operatorname{tr}(\Lambda_l^2) \{1 + o(1)\}, l = 1, 2.$$

Under H_0 , we have $\Lambda = \Lambda_1 = \Lambda_2$. A straightforward approach to estimate $p^{-1} \operatorname{tr}(\Lambda^2)$ is

$$p^{-1} \widehat{\operatorname{tr}(\Lambda^2)} = p(n_1 + n_2)^{-1} \{n_1 (A'_{n_1} - \hat{\delta}_{n_1}) + n_2 (B'_{n_2} - \hat{\delta}_{n_2})\},$$

where $A'_{n_1} = n_1^{-1}(n_1 - 1)^{-1} \sum_{i \neq i'} (\hat{\mathbf{u}}_i^T \hat{\mathbf{u}}_{i'})^2$, $B'_{n_2} = n_2(n_2 - 1)^{-1} \sum_{j \neq j'} (\hat{\mathbf{v}}_j^T \hat{\mathbf{v}}_{j'})^2$, and

$$\begin{aligned} \hat{\delta}_{n_l} = & \frac{1}{n_l^2} \left\{ 2 - \frac{2\hat{E}(R_{li}^{-2})}{\hat{E}^2(R_{li}^{-1})} + \frac{\hat{E}^2(R_{li}^{-2})}{\hat{E}^4(R_{li}^{-1})} \right\} \\ & + \frac{1}{n_l^3} \left\{ -\frac{6\hat{E}^2(R_{li}^{-2})}{\hat{E}^4(R_{li}^{-1})} + \frac{2\hat{E}(R_{li}^{-2})\hat{E}(R_{li}^{-3})}{\hat{E}^5(R_{li}^{-1})} + \frac{8\hat{E}(R_{li}^{-2})}{\hat{E}^2(R_{li}^{-1})} - \frac{2\hat{E}(R_{li}^{-3})}{\hat{E}^3(R_{li}^{-1})} \right\}, \end{aligned}$$

for $l = 1, 2$. By replacing $p^{-1} \operatorname{tr}(\Lambda^2)$ and $E(R_{li}^{-k})$ by $p^{-1} \widehat{\operatorname{tr}(\Lambda^2)}$ and $\hat{E}(R_{li}^{-k})$ in v_{0, n_1, n_2}^2 and δ_{n_1, n_2} , we have $\hat{v}_{0, n_1, n_2}^2$ and $\hat{\delta}_{n_1, n_2}$.

The next theorem shows that the estimator $\hat{v}_{0, n_1, n_2}^2$ is ratio consistent. The proof is given in the Appendix.

Theorem 3. Under Assumptions 1 and 2, the null hypothesis (2), and elliptically symmetric distributions, as $\min\{n_1, n_2\} \rightarrow \infty$ and $p \rightarrow \infty$, we have

$$\hat{v}_{0, n_1, n_2} / v_{0, n_1, n_2} \xrightarrow{P} 1,$$

where $\xrightarrow{P}$ denotes convergence in probability.

Remark 2. Then, by Slutsky's theorem, $\hat{v}_{0, n_1, n_2}^{-1} (T'_{n_1, n_2} - p\hat{\delta}_{n_1, n_2}) \xrightarrow{D} N(0, 1)$, as long as $\hat{v}_{0, n_1, n_2}^{-1} p(\hat{\delta}_{n_1, n_2} - \delta_{n_1, n_2}) = o_p(1)$. It can be easily shown that $n_l^{-1} \sum_{i=1}^{n_l} R_{li}^{-k} = E(R_{li}^{-k}) \{1 + O_p(n^{-1/2})\}$, for $l = 1, 2$. Then, we know that $v_{0, n_1, n_2}^{-1} p(\hat{\delta}_{n_1, n_2} - \delta_{n_1, n_2}) = o_p(1)$ is valid for

$p = o(n_l^{3/2})$ ($l = 1, 2$) when Σ_1 and Σ_2 have bounded eigenvalues. Details can be found in the Appendix.

By Theorems 2 and 3, as $\text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2 > 0$ under the alternative, then the acceptance region at the test level $100\alpha\%$ is constructed as follows:

$$\left\{ (\mathbf{x}_1, \dots, \mathbf{x}_{n_1}; \mathbf{y}_1, \dots, \mathbf{y}_{n_2}) : \hat{v}_{0,n_1,n_2}^{-1} (T'_{n_1,n_2} - p\hat{\delta}_{n_1,n_2}) \leq z_\alpha \right\},$$

where z_α is the upper- α quantile of $N(0, 1)$.

Then, we wish to study the power performance for $\{T'_{n_1,n_2} - p\delta_{n_1,n_2}\}$; the asymptotic power function of $\{T'_{n_1,n_2} - p\delta_{n_1,n_2}\}$ can be written as

$$\beta_{n_1,n_2}(\mathbf{S}_1, \mathbf{S}_2, \Lambda_1, \Lambda_2, \alpha) = \Phi \left\{ -\mathcal{L}_{n_1,n_2}(\Lambda_1, \Lambda_2)z_\alpha + p v_{n_1,n_2}^{-1} \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2 \right\},$$

where $\mathcal{L}_{n_1,n_2}(\Lambda_1, \Lambda_2) = v_{n_1,n_2}^{-1} \hat{v}_{0,n_1,n_2}$ and $\Phi(\cdot)$ denotes the cumulative probability distribution of $N(0, 1)$. The following theorem gives the consistency of the proposed test. The proof is given in the Appendix.

Theorem 4. Under Assumptions 1 and 2, as $\min\{n_1, n_2\} \rightarrow \infty, p \rightarrow \infty$, if $(n_1 + n_2) p \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$ has larger order than $\{p^{-1} \text{tr}(\Lambda_1^2) + p^{-1} \text{tr}(\Lambda_2^2)\}$, there is

$$\beta_{n_1,n_2}(\Lambda_1, \Lambda_2, \mathbf{S}_1, \mathbf{S}_2, \alpha) \rightarrow 1.$$

Moreover, if both Σ_1 and Σ_2 are positive definite and have bounded eigenvalues, then $\{p^{-1} \text{tr}(\Lambda_1^2) + p^{-1} \text{tr}(\Lambda_2^2)\} = O(1)$. Thus, Theorem 4 ensures that the proposed test is consistent only if $p^{-1} \text{tr}(\Lambda_1 - \Lambda_2)^2$ is not shrinking faster than n^{-1} , as $p \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$ has the same order as $p^{-1} \text{tr}(\Lambda_1 - \Lambda_2)^2$.

4 | SIMULATION STUDIES

In this section, we perform some simulation studies to evaluate the finite sample properties of our testing method for hypothesis (1) in the high-dimensional case. For comparisons, we also conduct the Bartlett adjusted likelihood ratio test (Flury, 1986), Wald test (Schott, 1999), PLRT (Xu et al., 2014), and Liu & Zheng test (Liu et al., 2014). With each method, the data are generated from the following two scenarios:

- *Scenario 1 (multivariate Gaussian distribution):* Samples $\mathbf{x}_1, \dots, \mathbf{x}_{n_1}$ are from $N_p(\mathbf{0}, \Sigma_1)$ and $\mathbf{y}_1, \dots, \mathbf{y}_{n_2}$ are from $N_p(\mathbf{0}, \Sigma_2)$;
- *Scenario 2 (multivariate t-distribution):* Samples $\mathbf{x}_1, \dots, \mathbf{x}_{n_1}$ are from $t_6(\mathbf{0}, \Sigma_1)$ and $\mathbf{y}_1, \dots, \mathbf{y}_{n_2}$ are from $t_6(\mathbf{0}, \Sigma_2)$.

It is noted that the above two distributions are elliptically distributed. We choose $\Sigma_1 = \mathbf{I}_p$ and $\Sigma_2 = c(\rho^{|i-j|})_{i,j=1}^p$ with $c = 1, 2, \rho = 0$ for computing empirical sizes and with different $\rho \neq 0$ for computing empirical power. The combinations of (n_1, n_2) are set to be (50, 100), (100, 100), and (100, 200). The nominal significant level is set as $\alpha = 5\%$, and all empirical test sizes and empirical power are computed based on 5,000 simulations.

Table 2 for Scenario 1 reports the simulation results of Gaussian data for $p = 40, 80$. With large p , both the Wald test and the Bartlett adjusted likelihood ratio test have large empirical sizes, which indicates these two methods no longer work for high-dimensional data. Besides, compared to the PLRT and LZ test, our proposed method also has good size performance. To study the robustness of our proposed test on non-normal data, we go on to perform Scenario 2 for

TABLE 2 Simulation results (in percentage) for Scenario 1 based on 5,000 independent replications

<i>p</i>	<i>n</i> ₁	<i>n</i> ₂	Empirical sizes					Empirical sizes				
			<i>c</i> = 1					<i>c</i> = 2				
			Wald	Bartlett	PLRT	LZ	<i>T</i> ' _{<i>n</i>₁,<i>n</i>₂}	Wald	Bartlett	PLRT	LZ	<i>T</i> ' _{<i>n</i>₁,<i>n</i>₂}
40	50	100	0.66	23.95	5.08	5.98	4.90	6.02	22.92	5.78	6.46	5.36
	100	100	1.18	5.95	5.72	6.68	5.48	5.27	6.25	5.62	6.83	5.20
	100	200	2.43	6.66	5.44	6.34	5.04	6.31	6.36	5.40	5.61	5.59
80	50	100	–	–	–	7.44	4.76	–	–	–	6.92	4.89
	100	100	0.05	61.60	6.66	6.66	4.70	14.93	60.83	6.36	7.28	4.87
	100	200	0.65	51.82	4.70	6.12	5.22	13.35	50.81	5.19	5.79	5.23

Note. PLRT = pseudo-likelihood ratio test.

TABLE 3 Simulation results (in percentage) for Scenario 2 based on 5,000 independent replications

<i>p</i>	<i>n</i> ₁	<i>n</i> ₂	Empirical sizes				Empirical sizes			
			<i>c</i> = 1				<i>c</i> = 2			
			Wald	PLRT	LZ	<i>T</i> ' _{<i>n</i>₁,<i>n</i>₂}	Wald	PLRT	LZ	<i>T</i> ' _{<i>n</i>₁,<i>n</i>₂}
40	50	100	98.66	98.86	87.44	5.42	99.79	95.08	82.58	5.75
	100	100	99.97	99.74	85.74	4.92	99.97	98.49	80.77	4.82
	100	200	100	100	99.86	5.64	100	99.95	99.43	5.65
80	50	100	–	–	27.10	4.86	–	–	25.70	5.01
	100	100	100	97.30	19.18	4.86	100	93.32	18.87	4.91
	100	200	100	100	100	5.34	100	100	99.97	5.11

Note. PLRT = pseudo-likelihood ratio test.

the multivariate *t*-distribution, and the simulation results are displayed in Table 3. All other tests, except our proposed test, have unsatisfactory performance since the empirical sizes are far away from the nominal level of 5%. As there is no existing method for large $p > \max(n_1, n_2)$, then we choose $p = 400, 800$ to handle our proposed method; we can see that the proposed method has satisfactory size performance for $\rho = 0$ and has nontrivial powers for a range of c and ρ under both a normal distribution and a *t*-distribution in Tables 4 and 5.

Figure 1 shows the density plot for the 2,000 observed values of the proposed method with $\rho = 0.0$ and $c = 2$. In both cases, $p < \min(n_1, n_2)$ and $p > \max(n_1, n_2)$, the normality result appears to be satisfied by the density plots for large (p, n_1, n_2) , which validates our theoretical asymptotic normality results; therefore, it is also a good evidence for why the empirical sizes of the proposed method perform very well.

TABLE 4 The performance of the proposed test under Scenario 1 with a range of ρ and c

<i>p</i>	<i>n</i> ₁	<i>n</i> ₂	$\rho = 0.0$		$\rho = 0.2$		$\rho = 0.4$	
			<i>c</i> = 1	2	<i>c</i> = 1	2	<i>c</i> = 1	2
400	50	100	5.06	4.63	34.89	34.83	100	100
	100	100	5.01	4.67	62.49	62.60	100	100
	100	200	5.11	4.87	83.42	83.56	100	100
800	50	100	4.73	4.89	34.83	35.02	100	100
	100	100	4.71	4.81	63.35	63.57	100	100
	100	200	5.02	5.27	84.21	83.33	100	100

TABLE 5 The performance of the proposed test under Scenario 2 with a range of ρ and c

p	n_1	n_2	$\rho = 0.0$		$\rho = 0.2$		$\rho = 0.4$	
			$c = 1$	2	$c = 1$	2	$c = 1$	2
400	50	100	4.65	4.80	35.09	36.06	100	100
	100	100	4.79	4.73	62.86	62.47	100	100
	100	200	4.96	4.65	84.15	83.56	100	100
800	50	100	4.97	4.88	36.75	36.37	100	100
	100	100	4.72	4.89	62.58	64.23	100	100
	100	200	5.37	5.05	84.34	84.86	100	100

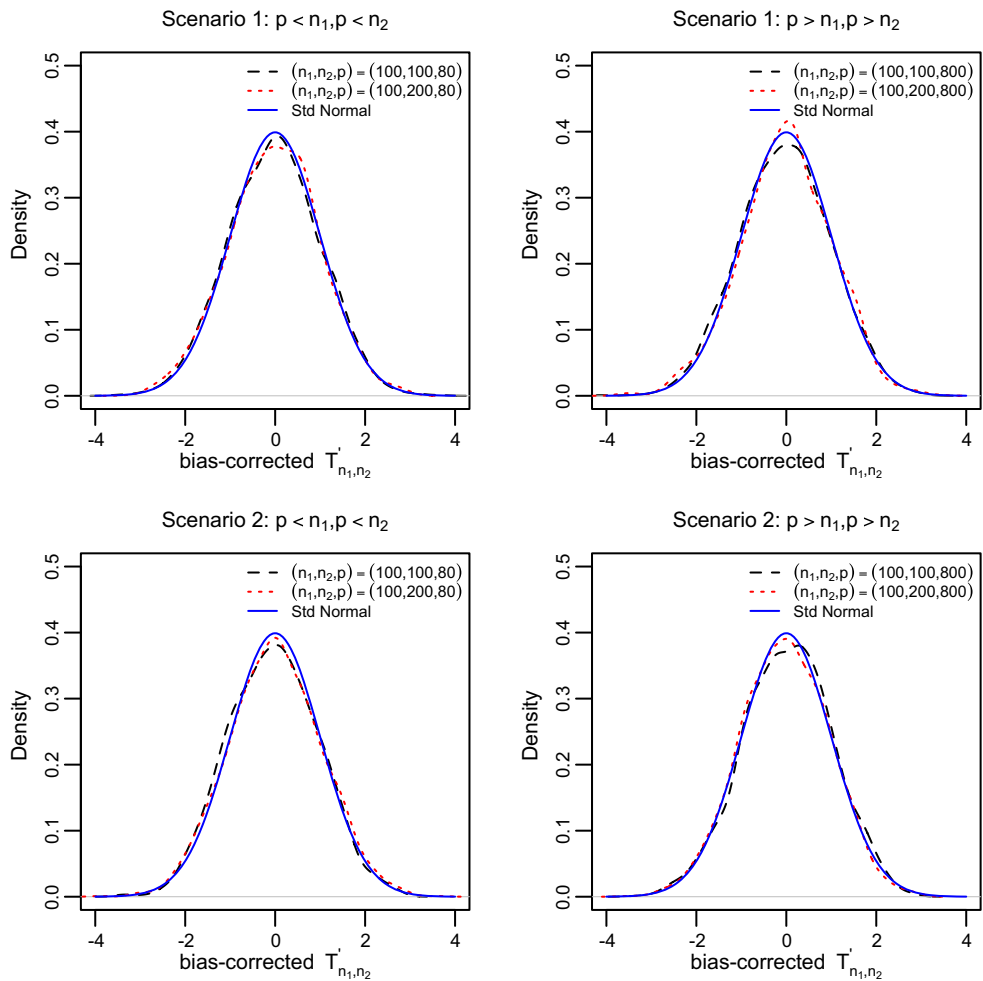


FIGURE 1 The null distributions of standardized and bias-corrected $\{T'_{n_1, n_2} - p\delta_{n_1, n_2}\}$ with a range of p , n_1 and n_2 for $c = 2$ [Colour figure can be viewed at wileyonlinelibrary.com]

For comparing empirical powers, as dimension p increases, both the Wald test and the Bartlett test lead to an increasing test size. Thus, we only compared with the LZ test, PLRT, and our proposed method. It can be seen that our proposed method has the greater powers although the

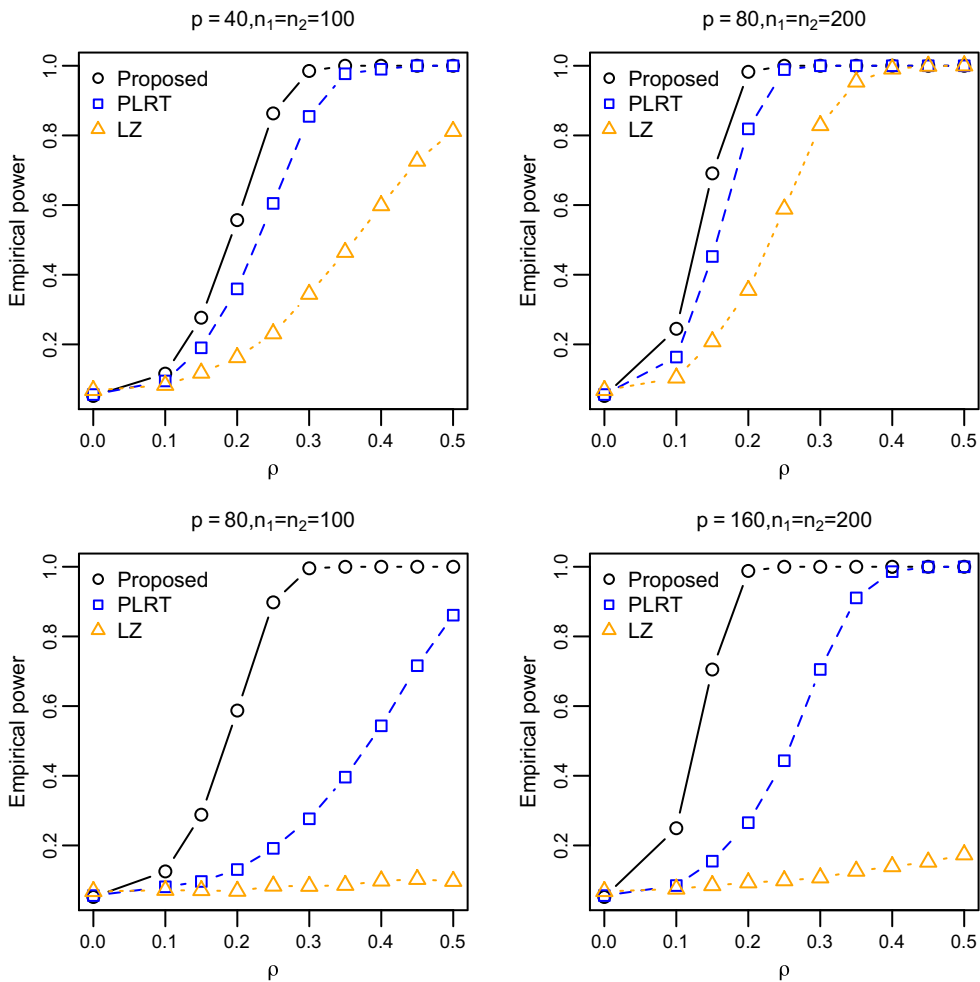


FIGURE 2 Power curve comparison between the pseudo-likelihood ratio test (PLRT), the LZ test, and the proposed procedure with different $p/n_1, p/n_2$ for a normal distribution with $c = 2$ [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/sjst.12350)]

empirical sizes of these three methods are all so close to the nominal level $\alpha = 0.05$ when the sample sizes become greater. Moreover, fixing the sample sizes n_1 and n_2 , the powers of the PLRT and LZ test decrease, but the powers of our proposed test seem to be stable when p becomes greater; these results are illustrated in Figure 2. All of these simulation results show that our proposed test is valid and robust, especially for heavy-tailed or skewed data.

5 | CONCLUSIONS AND DISCUSSIONS

This paper proposes a new nonparametric test based on a spatial sign covariance matrix for testing the proportionality of two high-dimensional covariance matrices. Its asymptotic normality is established under both null and alternative hypotheses. The advantage of our proposed test is valid for some heavy-tailed distributions, for instance, multivariate t -distributions with respect to the dimension much larger than the sample sizes. One limitation of our method is, as did in the

bias-correction procedure of Theorem 2 with $n = n_1 + n_2$, it requires that p is not growing faster than n^2 , that is, $p = O(n^2)$ where $n = n_1 + n_2$. In addition, when δ_{n_1, n_2} in $\{T'_{n_1, n_2} - p\delta_{n_1, n_2}\}$ is replaced by the estimator $\hat{\delta}_{n_1, n_2}$, the bias-corrected method is valid for $p = o(n^{3/2})$. In the future, we try to relax this limitation and extend our work to general cases.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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APPENDIX

TECHNICAL PROOFS

This section contains the main proofs of Theorems 1, 2, 3, and 4.

A.1 | Proof of Theorem 1

Suppose that $\psi_h = \psi'_h$, $h = 1, \dots, p$, that is,

$$E\left(\frac{\lambda_h Z_h^2}{\sum_{k=1}^p \lambda_k Z_k^2}\right) = E\left(\frac{\lambda'_h Z_h^2}{\sum_{k=1}^p \lambda'_k Z_k^2}\right), h = 1, \dots, p, \quad (\text{A1})$$

where $Z_1^2, \dots, Z_p^2$ are mutually independent χ^2 -variates with one degree of freedom 1.

1. Thus, if there exists some $h \in \{1, \dots, p\}$ such that $\lambda_h = 0$, then we have $\lambda'_h = 0$. Let $\rho_h = \lambda'_h / \lambda_h$ for some $\lambda_h \neq 0$ and $\lambda'_h \neq 0$. Denoting $\tilde{Z}_h^2 = \lambda_h Z_h^2$, then we have

$$E\left(\frac{\tilde{Z}_h^2}{\sum_{k=1}^p \tilde{Z}_k^2}\right) = E\left(\frac{\rho_h \tilde{Z}_h^2}{\sum_{k=1}^p \rho_k \tilde{Z}_k^2}\right), \quad 1 \leq h \leq p. \quad (\text{A2})$$

Let $\rho_1, \dots, \rho_p$ be ordered as $\rho_{(1)} \leq \dots \leq \rho_{(p)}$.

2. It is clear that if $\rho_{(1)} = \rho_{(p)}$, then $\lambda_1 / \lambda'_1 = \dots = \lambda_p / \lambda'_p$.

3. Suppose $\rho_{(1)} < \rho_{(p)}$ and k_0 satisfying $\rho_{k_0} = \rho_{(p)}$, then we have $\sum_{k=1}^p \rho_k \tilde{Z}_k^2 < \rho_{k_0} \sum_{k=1}^p \tilde{Z}_k^2$. Thus,

$$E\left(\frac{\rho_{k_0} \tilde{Z}_{k_0}^2}{\sum_{k=1}^p \rho_k \tilde{Z}_k^2}\right) > E\left(\frac{\rho_{k_0} \tilde{Z}_{k_0}^2}{\sum_{k=1}^p \rho_{k_0} \tilde{Z}_k^2}\right) = E\left(\frac{\tilde{Z}_{k_0}^2}{\sum_{k=1}^p \tilde{Z}_k^2}\right),$$

which contradicts (A2). Therefore, we have $\rho_{(1)} = \rho_{(p)}$, that is, $\lambda_1 / \lambda'_1 = \dots = \lambda_p / \lambda'_p$.

A.2 | Proof of Theorem 2

Define a sequence of random variables $\{\mathbf{z}_1, \dots, \mathbf{z}_{n_1+n_2}\}$, as follows:

$$\mathbf{z}_i = \mathbf{u}_i, 1 \leq i \leq n_1, \quad \text{and} \quad \mathbf{z}_{n_1+j} = \mathbf{v}_j, 1 \leq j \leq n_2.$$

Let $E_k(\cdot)$ denote the conditional expectation conditional on $\{\mathbf{z}_1, \dots, \mathbf{z}_k\}$. Define $D_{n,k} = p^{-1}\{E_k(T_{n_1, n_2}) - E_{k-1}(T_{n_1, n_2})\}$. Then, it is easy to show that $p^{-1}\{T_{n_1, n_2} - E(T_{n_1, n_2})\} = \sum_{k=1}^{n_1+n_2} D_{n,k}$.

Hence, the sequence $\{D_{n,1}, \dots, D_{n,n_1+n_2}\}$ constitutes a martingale difference with respect to the σ -fields $\sigma(\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_k)$. To derive the martingale central limit theorem, we need the following lemma.

Lemma 3. Under Assumption 2 and as $\min\{n_1, n_2\} \rightarrow \infty$, we have

$$p^2 \sum_{k=1}^{n_1+n_2} \sigma_{n,k}^2 / \text{Var}(T_{n_1,n_2}) \xrightarrow{p} 1 \quad \text{and} \quad \sum_{k=1}^{n_1+n_2} \mathbb{E}(D_{nk}^4) = p^{-4} o\{\text{Var}(T_{n_1,n_2})\},$$

$$\text{where } \sigma_{n,k}^2 = \mathbb{E}_{k-1}(D_{n,k}^2).$$

The above lemma shows the variance of martingale is convergent in probability, and the Lindeberg condition can be established. The proof is placed in the Supplementary Material.

In order to study the asymptotic normality of T'_{n_1,n_2} , as shown in the proof of Lemma 2, we have

$$\mathbb{E}(T'_{n_1,n_2}) = \mathbb{E}(T_{n_1,n_2}) + p\delta_{n_1,n_2} + o(pn^{-3}) + O(n^{-2}),$$

then we only need to study the asymptotic normality of the leading term in T_{n_1,n_2} .

In the Supplementary Material, we also prove that

$$\begin{aligned} \text{Var}(T_{n_1,n_2}) &= p^2 \{ \text{Var}(A_{n_1}) + \text{Var}(B_{n_2}) + 4\text{Var}(C_{n_1,n_2}) - 4\text{Cov}(A_{n_1}, C_{n_1,n_2}) - 4\text{Cov}(B_{n_2}, C_{n_1,n_2}) \} \\ &= \left\{ \frac{4}{n_1(n_1-1)} \frac{\text{tr}^2(\Lambda_1^2)}{(p+2)^2} + \frac{8}{n_1} \frac{p \text{tr}(\Lambda_1^4) - \text{tr}^2(\Lambda_1^2)}{p^2(p+2)} + \frac{4}{n_2(n_2-1)} \frac{\text{tr}^2(\Lambda_2^2)}{(p+2)^2} \right. \\ &\quad + \frac{8}{n_2} \frac{p \text{tr}(\Lambda_2^4) - \text{tr}^2(\Lambda_2^2)}{p^2(p+2)} - \frac{16}{n_1} \frac{p \text{tr}(\Lambda_1^3 \Lambda_2) - \text{tr}(\Lambda_1 \Lambda_2) \text{tr}(\Lambda_1^2)}{p^2(p+2)} \\ &\quad + \frac{8}{n_1 n_2} \frac{\text{tr}^2(\Lambda_1 \Lambda_2)}{(p+2)^2} + \left(\frac{8}{n_1} + \frac{8}{n_2} \right) \frac{p \text{tr}(\Lambda_1 \Lambda_2)^2 - \text{tr}^2(\Lambda_1 \Lambda_2)}{p^2(p+2)} \\ &\quad \left. - \frac{16}{n_2} \frac{p \text{tr}(\Lambda_2^3 \Lambda_1) - \text{tr}(\Lambda_1 \Lambda_2) \text{tr}(\Lambda_2^2)}{p^2(p+2)} \right\} (1 + o(1)). \end{aligned}$$

Following Lemma 3 and applying the martingale central limit theorem (Hall & Hyde, 1980), we conclude that

$$\frac{T_{n_1,n_2} - \mathbb{E}(T_{n_1,n_2})}{\text{Var}(T_{n_1,n_2})} \xrightarrow{d} N(0, 1).$$

A.3 | Proof of Theorem 3

Recall that $\mathbb{E}(pA'_{n_1}) = p \text{tr}(\mathbf{S}_1^2) + p\delta_{n_1} + o(n_1^{-1})$, as $\tilde{A}_{n_1} = A'_{n_1} - \delta_{n_1}$, then $\mathbb{E}(p\tilde{A}_{n_1}) = p \text{tr}(\mathbf{S}_1^2) + o(n_1^{-1})$. Notice that

$$\text{Var}(p^2 \tilde{A}_{n_1} / \text{tr}(\Lambda_1^2)) = O \left[\frac{p^4}{\text{tr}^2 \Lambda_1^2} \left\{ \frac{2}{n_1(n_1-1)} \left(\frac{3\text{tr}^2(\Lambda_1^2) + 6 \text{tr}(\Lambda_1^4)}{p^2(p+2)^2} - \frac{\text{tr}^2(\Lambda_1^2)}{p^4} \right) \right\} \right].$$

As $\text{tr}(\Lambda_1^4)/\text{tr}^2(\Lambda_1^2) \rightarrow 0$, hence $p^2 \tilde{A}_{n_1} / \text{tr}(\Lambda_1^2) \xrightarrow{p} 1$. Moreover, under H_0 , there are $\Lambda_1 = \Lambda_2 = \Lambda$, $p^2 \tilde{A}_{n_1} / \text{tr}(\Lambda^2) \xrightarrow{p} 1$. As

$$\hat{R}_{1i} - R_{1i} = \|\mathbf{x}_i - \hat{\boldsymbol{\mu}}_1\| - \|\mathbf{x}_i\| \leq \|\hat{\boldsymbol{\mu}}_1\| = O_p(c_1^{-1} n_1^{-1/2}),$$

we have

$$\hat{R}_{1i}^{-1} - R_{1i}^{-1} = E(R_{1i}^{-1}) \left(O_p \left(n_1^{-1/2} \right) \right), \quad n_1^{-1} \sum_{i=1}^{n_1} \hat{R}_{1i}^{-1} = n_1^{-1} \sum_{i=1}^{n_1} R_{1i}^{-1} \{1 + O_p(n_1^{-1/2})\}.$$

Moreover, we have $n_1^{-1} \sum_{i=1}^{n_1} R_{1i}^{-k} = E(R_{1i}^{-k}) \left(1 + O_p \left(n_1^{-1/2} \right) \right)$ by the assumption $E^2(R_{1i}^{-k}) / \{E(R_{1i}^{-k})\}^2 = O(1)$. Thus, we have

$$n_1 \sum_{i=1}^{n_1} \hat{R}_{1i}^{-2} / \left(\sum_{i=1}^{n_1} \hat{R}_{1i}^{-1} \right)^2 = E(R_{1i}^{-2}) / \{E(R_{1i}^{-1})\}^2 \{1 + O_p(n_1^{-1/2})\}$$

and

$$n_1^2 \sum_{i=1}^{n_1} \hat{R}_{1i}^{-3} / \left(\sum_{i=1}^{n_1} \hat{R}_{1i}^{-1} \right)^3 = E(R_{1i}^{-3}) / \{E(R_{1i}^{-1})\}^3 \{1 + o_p(1)\}.$$

Therefore, $\hat{\delta}_{n_1}$ converges to δ_{n_1} in probability and $p\hat{\delta}_{n_1} = p\delta_{n_1} \{1 + o_p(1)\}$ with $p = O(n_1^2)$. Then, $p^2 \tilde{A}'_{n_1} / \text{tr}(\Lambda^2) \xrightarrow{P} 1$, where $\tilde{A}'_{n_1} = A'_{n_1} - \hat{\delta}_{n_1}$. Similarly, we have $p^2 \tilde{B}'_{n_2} / \text{tr}(\Lambda^2) \xrightarrow{P} 1$. Hence, we proved that $\hat{v}_{0,n_1,n_2} / v_{0,n_1,n_2} \xrightarrow{P} 1$.

Moreover, when $\hat{v}_{0,n_1,n_2}^{-1} p(\hat{\delta}_{n_1,n_2} - \delta_{n_1,n_2}) = o_p(1)$, we have $\hat{v}_{0,n_1,n_2}^{-1} (T'_{n_1,n_2} - p\hat{\delta}_{n_1,n_2}) \xrightarrow{D} N(0, 1)$; note that $v_{0,n_1,n_2}^{-1} p(\hat{\delta}_{n_1,n_2} - \delta_{n_1,n_2}) = O(np\delta_{n_1,n_2} n^{-1/2})$. Thus, the test is valid for using $\hat{\delta}_{n_1,n_2}$ instead of δ_{n_1,n_2} in $\{T'_{n_1,n_2} - p\delta_{n_1,n_2}\}$ with $p = o(n^{3/2})$.

A.4 | Proof of Theorem 4

First, we will show that $\mathcal{L}_{n_1,n_2}(\Lambda_1, \Lambda_2)$ is bounded. In fact, $\hat{v}_{0,n_1,n_2}$ is a ratio-consistent estimator of $2p^{-1}n_1^{-1} \text{tr}(\Lambda_1^2) + 2p^{-1}n_1^{-1} \text{tr}(\Lambda_2^2)$. Hence, we only need to show that $v_{n_1,n_2}^{-1} [2p^{-1}n_1^{-1} \text{tr}(\Lambda_1^2) + 2p^{-1}n_1^{-1} \text{tr}(\Lambda_2^2)]$ is bounded. Notice that $v_{n_1,n_2}^2 \geq 4n_1^{-2}p^{-2} \text{tr}^2(\Lambda_1^2) + 4n_2^{-2}p^{-2} \text{tr}^2(\Lambda_2^2)$. Then, we have

$$\mathcal{L}_{n_1,n_2}(\Lambda_1, \Lambda_2) \leq \frac{n_2^{-1} \text{tr}(\Lambda_1^2) + n_1^{-1} \text{tr}(\Lambda_2^2)}{\sqrt{n_1^{-2} \text{tr}^2(\Lambda_1^2) + n_2^{-2} \text{tr}^2(\Lambda_2^2)}}.$$

Let $u_0 = \text{tr}(\Lambda_1^2) / \text{tr}(\Lambda_2^2)$. The right-hand side in the above inequality can be written as a function of u_0 , for instance, $h(u_0)$. It is easy to obtain that $h(u_0)$ is maximized at $u_0 = k_n / (1 - k_n)$, where $k_n = n_1 / (n_1 + n_2)$. Then, we have

$$\beta_{n_1,n_2}(\Lambda_1, \Lambda_2, \mathbf{S}_1, \mathbf{S}_2, \alpha) \geq \Phi \left(-z_\alpha k_n^{-1} (1 - k_n)^{-1} + p v_{n_1,n_2}^{-1} \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2 \right),$$

where the right-hand side indicates the lower bound of the power function and implies that the test is powerful when $p \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$ is at the same or a larger order of v_{n_1,n_2} .

Under Assumption 2, as $k_n \rightarrow \eta_1$, to prove the consistency of the proposed test, it is only enough to prove $p v_{n_1,n_2}^{-1} \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$ tending to infinity when both n and p tend to infinity. By the Cauchy-Schwarz inequality, we note that the variance of $\{T'_{n_1,n_2} - p\delta_{n_1,n_2}\}$ can be dominated by

$$4p^{-2} \{n_1^{-1} \text{tr}(\Lambda_1^2) + n_2^{-1} \text{tr}(\Lambda_2^2)\}^2 + 8p^{-2} \{n_1^{-1} \text{tr}(\Lambda_1^2) + n_2^{-1} \text{tr}(\Lambda_2^2)\} \text{tr}(\Lambda_1 - \Lambda_2)^2.$$

Thus, the test statistic $\{T'_{n_1,n_2} - p\delta_{n_1,n_2}\}$ has nontrivial power $p \text{tr}(\mathbf{S}_1 - \mathbf{S}_2)^2$ at the least order of $n_1^{-1} p^{-1} \text{tr}(\Lambda_1^2) + n_2^{-1} p^{-1} \text{tr}(\Lambda_2^2)$.